

Variable Selection and the Interpretation of Principal Subspaces

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Principal component analysis is widely used in the analysis of multivariate data in the agricultural, biological, and environmental sciences. The first few principal components (PCs) of a set of variables are derived variables with optimal properties in terms of approximating the original variables. This paper considers the problem of identifying subsets of variables that best approximate the full set of variables or their first few PCs, thus stressing dimensionality reduction in terms of the original variables rather than in terms of derived variables (PCs) whose definition requires all the original variables. Criteria for selecting variables are often ill defined and may produce inappropriate subsets. Indicators of the performance of different subsets of the variables are discussed and two criteria are defined. These criteria are used in stepwise selection-type algorithms to choose good subsets. Examples are given that show, among other things, that the selection of variable subsets should not be based only on the PC loadings of the variables.

Key Words: Loadings; Multiple regression; Principal components; Reification; Stepwise selection.

1. INTRODUCTION

Principal component analysis (PCA) is widely used throughout science as a dimension-reducing tool. The Current Index in Statistics identifies over 3,000 articles in the period 1995–1998 with the phrase ‘Principal Component Analysis’ or ‘Principal Components Analysis’ in their title, abstract, or keywords. A small sample of recent applications in the agricultural, biological, and environmental sciences is Teitelman and Eeckman (1996), Villar, Garcia, Iglesias, Garcia, and Otero (1996), Durrieu et al. (1997), Baeriswyl and Rebetez (1997), Yu et al. (1998), and Ferraz, Esposito, Bruns, and Duran (1998). When PCA is done on a large number of variables, the results may not be easy to interpret. Meaningful

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interpretations can result from identifying a small subset of variables that approximate the individual principal components (PCs) or the subspaces spanned by groups of PCs. Attempts to tackle this variable reduction problem date back at least as far as Jolliffe (1972, 1973). At the same time, this approach suggests that dimensionality reduction of a data set with many variables may be sought directly in terms of subsets of variables rather than through linear combinations (PCs) of the variables.

A motivating example is taken from Somers (1986) (see that article for greater detail). The example consists of measurements (in millimeters) of 13 morphometric variables for 63 crayfish collected in Lake Opeongo, Ontario. The 13 variables, x_1 to x_{13} , are, respectively, carapace length, tail length, carapace width, carapace depth, tail width, areola length, areola width, rostrum length, rostrum width, postorbital width, propodus length, propodus width, and dactyl length. It is common to base studies of morphometric data on the PCA of the covariance matrix of the log-transformed data (see Jolicoeur [1963] for the original suggestion to this effect), and we follow this convention.

The (unit-norm) loadings for the first five PCs generated by this covariance matrix, together with the percentage variance and cumulative percentage variance associated with those PCs, are given in Table 1.

The first PC accounts for 63.71% of the log-transformed data set’s total variability, and the first two PCs together account for 79.92%. Both of these components have substantial loadings on several of the variables, and neither is straightforward to interpret. A question that then arises is whether a small number of variables (ideally two) could be almost as informative as these two components. When dealing with the problem of interpreting PCs in terms of a subset of variables, there are a number of facets, including (i) which principal components are of interest, (ii) how many variables to keep, (iii) whether the components of interest are to be considered one at a time or simultaneously, (iv) how the variables are to be selected, and (v) what is meant by ‘approximating’ the components.

Table 1. Loadings and Percentage Variance Accounted for by the First Five PCs in the Crayfish Data

Variable	PC1	PC2	PC3	PC4	PC5
x_1	− 0.166	0.152	0.076	0.018	− 0.045
x_2	− 0.105	0.417	0.134	0.081	0.112
x_3	− 0.217	0.179	0.125	− 0.006	− 0.033
x_4	− 0.195	0.135	0.163	− 0.007	− 0.018
x_5	− 0.088	0.559	0.117	0.070	0.153
x_6	− 0.223	0.186	0.018	0.011	− 0.885
x_7	− 0.271	0.247	− 0.639	− 0.649	0.100
x_8	− 0.139	0.248	0.124	0.154	0.378
x_9	− 0.166	0.119	0.103	− 0.079	0.038
x_{10}	− 0.175	0.178	0.207	0.115	− 0.047
x_{11}	− 0.451	− 0.275	− 0.007	0.047	0.062
x_{12}	− 0.458	− 0.363	0.499	− 0.415	0.101
x_{13}	− 0.505	− 0.173	− 0.441	0.591	0.076
% Variance	63.71	16.21	7.41	4.18	2.62
Cumulative % variance	63.71	79.92	87.32	91.50	94.12

Although other choices are possible (see Jolliffe 1986, sec. 10.1; Jolliffe 1989), the principal components of interest in (i) are conventionally those with the largest variance. How many of these to keep can be determined in a variety of ways—Jolliffe (1986, sec. 6.1) and Richman (1992) review some of many of the methods that have been suggested. For the purposes of this paper, we will assume that the choice has already been made.

Aspect (ii) depends on the choices made for (i) and (v). With regard to (iii), most previous work has concentrated on looking at each component separately, whereas the present paper deals with them simultaneously. We see two advantages to this approach. First, if we are retaining q PCs, we are often interested in interpreting the space spanned by those PCs rather than being wedded to individual PCs. This is the reasoning behind rotation in PCA or factor analysis. A second advantage is that we do not need to worry about different subsets of variables being best for different PCs, with the possibility that the union of these subsets may lead to a much larger subset of the original variables than is strictly necessary for a joint interpretation. An example of this will be seen in Section 4.2, where Jeffers' pitprop data set is discussed.

Turning to (iv), there are a variety of ways of choosing a subset. Many are based on intuition rather than on a well-defined criterion. For example, it is common practice to simplify a PC by ignoring variables whose loadings are small for that component. Cadima and Jolliffe (1995) showed that this approach is potentially misleading when one component at a time is considered. The examples later in this paper illustrate that similar problems occur when q components are considered simultaneously.

We firmly believe that the choice of variables should be mainly guided by one or more well-defined criteria rather than intuition, and this leads us to facet (v). It is difficult to envisage criteria that explicitly define interpretability, although rotation methods attempt to do so with their definition of simplicity. Hence, our aim in this paper is to define and illustrate two criteria that measure rigorously how closely we have approximated the retained components. The assured reduction in dimensionality, due to retaining only a subset of variables, means that difficulties in interpretation become less likely, but improved interpretability in an intuitive sense cannot be guaranteed. However, the examples illustrate the gains that may be possible.

The existence of meaningful and clearly defined criteria that are to be optimized in order to ensure the best possible approximation in some relevant sense is crucial to the discussion of the variable selection problem. Some authors (Gonzalez, Evry, Cl  roux, and Rioux 1990) have hinted at the fact that the absence of such well-known criteria has hindered the study of the variable-selection problem in PCA. Clearly defined optimization criteria are also essential to bypass the other problem that has affected the variable-selection problem, i.e., the exponential growth in the number of possible subsets of the p original variables, with the implication that a full search of all possible subsets becomes infeasible for any data set with even a fair number of variables. With a clearly defined optimization criterion, it will be possible to use the standard forward selection, backward elimination, and stepwise selection strategies that, without ensuring that the optimum k -variable subset will necessarily be found, should at least provide reasonable answers.

In the single component case, there is an obvious answer to aspect (v), using multiple correlation as a criterion, but things become more complicated in the simultaneous case. Jolliffe (1973) adopted a fairly ad hoc approach to the variable selection problem. None of his methods for variable selection attempted a global optimization of a well-defined criterion, and as a measure of how well a subset of variables represented a set of q components, he used the measure

$$Q_1 = \sum_{i=1}^q \lambda_i r_i / \sum_{i=1}^q \lambda_i,$$

where λ_i is the variance of the i th original component and r_i is the maximum correlation between the i th component and a component derived from the reduced set of variables. Q_2 , where r_i is replaced by rank correlations, was also used.

McCabe (1986) adopted a more rigorous approach to the problem. In an earlier paper (McCabe 1984), he introduced the idea of principal variables, which were an alternative to principal components. Principal variables are subsets of the variables that satisfy various optimality criteria, these criteria being parallel to those satisfied by principal components. Different criteria are proposed, leading to four separate definitions of principal variables. In McCabe (1986), one of these definitions is considered and the principal variables so derived are used to predict a subset of principal components.

Other authors have differing ideas of what is meant by approximating the principal components. Krzanowski (1987) is interested in structure, such as clusters of observations in the subspace of the first few components, and therefore compares the projections of the observations onto subspaces spanned by the first few original and reduced components using Procrustes rotation (see also Jolliffe 1987). Bonifas, Escoufier, Gonzales, and Sabatier (1984) suggest finding a subset of variables that maximizes the so-called RV-coefficient between matrices based on the original variables and on the subset. Gonzalez et al. (1990) pick up on the idea and propose a search algorithm to identify the optimum k -variable subsets. Falguerolles and Jmel (1993) propose a choice of variables based on the results of fitting a graphical model linking the variables.

In the present paper, we suggest that there are two main approaches to problem (v) that make sense. These result in two fairly simple indicators for proximity, namely,

- (a) the subspace spanned by the components of interest is close to the subspace spanned by the chosen subset of variables or
- (b) the components of interest provide an approximation to the full original data set that is similar to that obtained using the chosen subset of variables.

Section 2 gives precise definitions of what we mean by (a) and (b) and explores the properties of these criteria. We also discuss the connections between these criteria and those of earlier authors. Section 3 discusses algorithms for finding good subsets once a criterion has been chosen. In Section 4, the crayfish example above is revisited, and the ideas from Sections 2 and 3 are applied to it. The classic pitprop example (Jeffers 1967) is also examined. It demonstrates clearly that simple interpretations based on ignoring

variables with near-zero loadings can, as in the single-component case, be misleading. A third, larger example from agriculture is also discussed. The examples show that it is often possible to approximate well the optimal performance of the first few PCs using the same number of variables as components. Standard stepwise selection algorithms, using the indicators from Section 2 as the criteria for inclusion/exclusion of variables in the subset, produce fairly good results, even for the larger example. Finally, Section 5 contains some concluding points and discussion.

2. CRITERIA FOR APPROXIMATION

Let us assume that we are concerned with a set of p original variables, which can be identified by a variable number (from 1 to p) fixed in advance. Assume also that the PCs of those variables are labeled $1, 2, \dots, p$ in descending order of eigenvalues. We are interested in a subset of q PCs, which will often be the first q but which can be any subset indexed by a subset $\mathcal{G}$ of q integers in the set $\{1, 2, \dots, p\}$. The subspace spanned by those PCs will be represented by G . We must also consider a subset $\mathcal{K}$ of k variable numbers to identify the subset of the original variables being considered for the approximation to those PCs (it is not necessary to assume that $k = q$). The subspace spanned by the k variables will be represented by K .

It will clarify ideas if we identify the p original variables with their n -dimensional vectors of observations. The $n \times p$ data matrix X , assumed (without loss of generality) to have been column-centered and to have rank p , defines the covariance matrix $S = (1/n)X'X$. Given the spectral decomposition $S = A\Lambda A'$, the p PCs of the data set are the columns of the $n \times p$ matrix XA . The subset of q PCs is given by the $n \times q$ matrix $XA_{\mathcal{G}}$, where $A_{\mathcal{G}}$ is the $p \times q$ submatrix of the matrix A of PC loadings (eigenvectors of S), which is obtained by retaining the q columns of A whose column numbers are in set $\mathcal{G}$. Likewise, the subset of k variables is given by the $n \times q$ matrix $XI_{\mathcal{K}}$, where $I_{\mathcal{K}}$ is the $p \times q$ submatrix of the $p \times p$ identity matrix I , which is obtained by retaining the k columns of I whose column numbers are in $\mathcal{K}$.

A key concept is that of the matrices of orthogonal projections onto the subspaces G and K . The first of these matrices is

$$P_G = XA_{\mathcal{G}}(A'_{\mathcal{G}}X'XA_{\mathcal{G}})^{-1}A'_{\mathcal{G}}X' = \frac{1}{n}XA_{\mathcal{G}}\Lambda_{\mathcal{G}}^{-1}A'_{\mathcal{G}}X' = \frac{1}{n}XS_{\{\mathcal{G}\}}^{-1}X', \quad (2.1)$$

where $S_{\{\mathcal{G}\}} = A_{\mathcal{G}}\Lambda_{\mathcal{G}}A'_{\mathcal{G}}$ is the $p \times p$ matrix that results from retaining only the contribution of those eigenvectors of the spectral decomposition of S whose corresponding eigenvalues have indices in $\mathcal{G}$ (which, in the case of the first q PCs, means the best rank q approximation to S) and $S_{\{\mathcal{G}\}}^{-1} = A_{\mathcal{G}}\Lambda_{\mathcal{G}}^{-1}A'_{\mathcal{G}}$ is its Moore–Penrose generalized inverse (Ramsay and Silverman 1997, p. 286).

The matrix of orthogonal projections onto the subspace K is

$$P_K = XI_{\mathcal{K}}(I'_{\mathcal{K}}X'XI_{\mathcal{K}})^{-1}I'_{\mathcal{K}}X' = \frac{1}{n}XI_{\mathcal{K}}S_{\mathcal{K}}^{-1}I'_{\mathcal{K}}X', \quad (2.2)$$

where $S_K = (1/n)I'_K X' X I_K$ is the $k \times k$ submatrix of the data covariance matrix S that results from retaining the k rows/columns whose row/column numbers are in K .

2.1 SUBSPACE SIMILARITY

An indicator of subspace similarity is Yanai's generalized coefficient of determination (GCD) (Ramsay, ten Berge, and Styan 1984). The same idea is discussed independently by Krzanowski (1988) when an orthogonal basis for each subspace is known. Yanai's GCD is also intimately related to the distance between subspaces discussed by Golub and Van Loan (1996, sec. 2.6.3). The GCD is essentially the matrix correlation of the matrices of orthogonal projections onto the two subspaces and can be interpreted as the average of the squared canonical correlations between two sets of variables spanning those subspaces (Ramsay et al. 1984). This definition, which can also be viewed as the cosine of the angle defined by the orthogonal projection matrices in the inner product space of $n \times n$ matrices with the matrix inner product $\langle A, B \rangle = \text{tr}(A'B)$ and the resulting matrix norm $\|A\| = (\langle A, A \rangle)^{1/2} = (\text{tr}(A'A))^{1/2}$, gives

$$\text{corr}(A, B) = \frac{\langle A, B \rangle}{\|A\| \cdot \|B\|} = \frac{\text{tr}(A'B)}{\sqrt{\text{tr}(A'A) \cdot \text{tr}(B'B)}} \quad (2.3)$$

and does not necessarily require that the two subspaces have the same dimension. Due to the Cauchy–Schwarz inequality, the quantity defined in (2.3) can only take values between -1 and 1 . In our case, we are talking about subspaces G and K of $\mathbb{R}^n$, and we have

$$\text{GCD}(G, K) = \text{corr}(P_G, P_K) = \frac{\text{tr}(P_G P_K)}{\sqrt{qk}}. \quad (2.4)$$

Using expressions (2.1) and (2.2), we have further that

$$\text{GCD}(G, K) = \text{tr} \left(SS_{\{G\}}^{-1} S I_K S_K^{-1} I'_K \right) / \sqrt{qk} = \text{tr} (S_{\{G\}} I_K S_K^{-1} I'_K) / \sqrt{qk}$$

since $SS_{\{G\}}^{-1} S = S_{\{G\}}$ (see Appendix A). Thus, $\text{GCD}(G, K) = \text{tr}([S_{\{G\}}]_{(K)} S_K^{-1}) / (qk)^{1/2}$, where $[S_{\{G\}}]_{(K)}$ is the $k \times k$ submatrix of $S_{\{G\}}$ obtained by retaining only the rows/columns with row/column number in K . In other words, denoting by a_i the eigenvectors of S (columns of A) and by a_i^K the subvectors of a_i that result from retaining only those elements in positions given in the set K , we have $[S_{\{G\}}]_{(K)} = \sum_{i \in G} \lambda_i a_i^K a_i^{K'}$. The expression for the GCD can therefore be rewritten as

$$\text{GCD}(G, K) = \frac{1}{\sqrt{qk}} \sum_{i \in G} \lambda_i a_i^{K'} S_K^{-1} a_i^K = \frac{1}{\sqrt{qk}} \sum_{i \in G} (r_m)_i^2, \quad (2.5)$$

where $(r_m)_i = (\lambda_i)^{1/2} (a_i^{K'} S_K^{-1} a_i^K)^{1/2}$ is the multiple correlation between the data set's i th PC and the k variables spanning K (see Cadima and Jolliffe 1995). Thus, the GCD for the subspaces has values between zero (if the subspaces are mutually orthogonal) and one (if $q = k$ and all q PCs are in K , i.e., if the subspaces coincide). It can be seen that the GCD's value is essentially determined by the magnitude of each PC's multiple correlation with the k variables (a reflection of the PCs' orthogonality properties), so very large values of the GCD can only be obtained with good fits for all q PCs.

2.2 SUBSETS OF VARIABLES AS PREDICTORS

As is well known, when the full $n \times p$ data matrix is orthogonally projected onto the subspace G spanned by a subset $\mathcal{G}$ of q PCs (i.e., fitted by linear regression on the q PCs), the resulting matrix is

$$P_G X = X S_{\{\mathcal{G}\}}^{-1} S = X A_{\mathcal{G}} A_{\mathcal{G}}' \quad (2.6)$$

(see Appendix A). The matrix correlation (2.3) between X and its projection on the q PCs is

$$\text{corr}(X, P_G X) = \frac{\langle X, P_G X \rangle}{\|X\| \cdot \|P_G X\|} = \frac{\|P_G X\|}{\|X\|} = \sqrt{\frac{\text{tr}(A_{\mathcal{G}}' S A_{\mathcal{G}})}{\text{tr}(S)}} = \sqrt{\frac{\sum_{i \in \mathcal{G}} \lambda_i}{\sum_{j=1}^p \lambda_j}},$$

i.e., the square root of the well-known “percentage of total variance accounted for” by the q PCs. By the properties of PCA, this correlation must represent an optimum over all matrices of orthogonal projections onto q -dimensional subspaces when $\mathcal{G}$ is the set of the first q PCs.

If, on the other hand, the full data matrix is orthogonally projected onto the subspace K spanned by a subset $\mathcal{K}$ of k variables (fitted by linear regression on the k variables), the resulting matrix is

$$P_K X = X I_{\mathcal{K}} S_{\mathcal{K}}^{-1} I_{\mathcal{K}}' S \quad (2.7)$$

and its matrix correlation with X is

$$\text{corr}(X, P_K X) = \sqrt{\frac{\text{tr}(X' P_K X)}{\text{tr}(X' X)}} = \sqrt{\frac{\text{tr}(S I_{\mathcal{K}} S_{\mathcal{K}}^{-1} I_{\mathcal{K}}' S)}{\text{tr}(S)}} = \sqrt{\frac{\text{tr}([S^2]_{(\mathcal{K})} S_{\mathcal{K}}^{-1})}{\text{tr}(S)}},$$

where $[S^2]_{(\mathcal{K})}$ is the $k \times k$ submatrix of S^2 obtained by retaining the k rows/columns associated with set $\mathcal{K}$. Now $[S^2]_{(\mathcal{K})} = \sum_{i=1}^p \lambda_i^2 a_i^{\mathcal{K}} a_i^{\mathcal{K}'}; \text{ hence,}$

$$\text{tr}([S^2]_{(\mathcal{K})} S_{\mathcal{K}}^{-1}) = \sum_{i=1}^p \lambda_i^2 a_i^{\mathcal{K}'} S_{\mathcal{K}}^{-1} a_i^{\mathcal{K}} = \sum_{i=1}^p \lambda_i (r_m)_i^2.$$

And so

$$r_m = \text{corr}(X, P_K X) = \sqrt{\frac{\sum_{i=1}^p \lambda_i (r_m)_i^2}{\sum_{j=1}^p \lambda_j}}. \quad (2.8)$$

It can be shown that the square of indicator (2.8) can be interpreted as the percentage of total variance accounted for by the k variable subset. The maximization of $\text{corr}(X, P_K X)$ therefore selects the k -variable subset that maximizes the same criterion (variance) as PCA, though here we are restricted to subsets of the observed variables rather than subsets of all linear combinations of those variables. This corresponds to the second of McCabe’s four criteria (see Appendix B) for principal variables. The ratio

$$\frac{\text{corr}(X, P_K X)}{\text{corr}(X, P_G X)} = \sqrt{\frac{\sum_{i=1}^p \lambda_i (r_m)_i^2}{\sum_{j \in \mathcal{G}} \lambda_j}} = \frac{\|P_K X\|}{\|P_G X\|} \quad (2.9)$$

will tell us how much worse the k -variable subset $\mathcal{K}$ performs in approximating the full data matrix X when compared with the q -PC subset $\mathcal{G}$. A direct comparison of both projected

matrices, i.e.,

$$\text{corr}(\mathbf{P}_G \mathbf{X}, \mathbf{P}_K \mathbf{X}) = \frac{\sum_{i \in \mathcal{G}} \lambda_i (r_m)_i^2}{\sqrt{\left(\sum_{i \in \mathcal{G}} \lambda_i \right) \left(\sum_{i=1}^p \lambda_i (r_m)_i^2 \right)}},$$

in the spirit of Krzanowski's (1987) method, could also be considered but would lose the reference point given by the original data matrix $\mathbf{X}$.

It should be noted that the r_m indicator (2.8), like the GCD indicator of subspace similarity (2.5), involves sums of squared multiple correlations between each PC and the k -variable subset $\mathcal{K}$ except that the GCD indicator involves only some PCs and the r_m indicator involves a weighted sum of all PCs, in which each PC's weight is its proportion of variance accounted for. Hence, the r_m indicator will be less demanding of good fits for low-eigenvalue PCs than for high-eigenvalue PCs. In addition, the r_m indicator can be viewed as the matrix correlation between two $n \times p$ data matrices (the second of which is of rank $k < p$) and will implicitly compare two sets of PCs, those of the original data matrix $\mathbf{X}$ and those of the projected data matrix $\mathbf{P}_K \mathbf{X}$. The closer are the set of eigenvectors and the relative eigenvalues of one matrix to those of the other matrix, the larger will be the matrix correlation.

3. ALGORITHMS FOR FINDING GOOD SUBSETS

Once a criterion, such as (2.5) or (2.8), has been chosen, the next question is how to find a subset of variables that does well, or preferably best, with respect to that criterion. If p is not too large, it is feasible to search all possible subsets of any given size k and hence guarantee that the best subset is discovered. For example, with 13 variables, the number of all possible proper subsets is 8,190, while with 20 variables, it grows to just over 1 million. With enough resources, it may be possible to do a complete search for all subsets of data sets with as many as 30 variables. However, for larger values of p (such as in example 4.3), it becomes infeasible to explore all subsets, and the need arises for an algorithm that only searches a fraction of the available subsets but that will usually include optimal or near-optimal subsets within its search.

There are clear parallels here with variable selection in multiple regression, and any of the algorithms used in that context can be used here. The best known are forward selection, stepwise selection, and backward elimination (cf., Neter, Wasserman, and Kutner 1990, sec. 12.4). None of these guarantee to find the best subset except when $k = 1$ (forward and stepwise selection) or $k = p - 1$ (backward elimination), and attempts have been made to devise other algorithms that reduce the chances of missing the best subset. For example, Gonzalez et al. (1990) suggest an algorithm when the optimality criterion is the RV-coefficient. We examine the behavior of these algorithms in the examples that follow.

4. EXAMPLES

4.1 THE CRAYFISH DATA

We saw earlier (Table 1) that the first two principal components of the covariance matrix of the log-transformed data account for 79.92% of the total variation in these transformed data. Examining all subsets of two (log-transformed) variables, we find that the combination of variables 5 (tail width) and 11 (propodus length) is the optimum choice for both indicators (2.5) and (2.8). With respect to r_m , this subset accounts for 75.63% of the total variability, which is 94.63% of the optimum for two-dimensional subspaces. The sum of the variances of (log-transformed) variables 5 and 11 account for only 21% of the total variance in all 13 variables, but when the data are projected onto the subspace spanned by these two variables, nearly 76% of the data set's variability is accounted for. The GCD between the subspaces spanned by the first two PCs and variables 5 and 11 is 0.9225. The practical significance of these values can be illustrated by comparing three possible scatterplots of the 63 crayfish, as is done in Figure 1, which gives the scatterplot on the first two PCs of the full (log-transformed) data set, the scatterplot on the only two PCs of the projected data matrix $P_K X$, which is obtained by regressing all 13 variables on variables 5 and 11, and the scatterplot on variables 5 and 11 only. The two rightmost plots in Figure 1 are on the same subspace (the subspace spanned by variables 5 and 11). The leftmost plot in Figure 1 is on a subspace that is close to the subspace spanned by variables 5 and 11, as measured by the $GCD = 0.9225$, and in this sense, the two-dimensional principal subspace can be interpreted as essentially the subspace spanned by tail width and propodus length. The center plot illustrates how the $n \times p$ data matrix obtained by replacing the observations in each variable with their estimates, as given by a regression on variables 5 and 11, is close to the original data matrix, as measured by the value of the r_m indicator, 0.8697. It depicts the scatter of points on the two-dimensional subspace spanned by variables 5 and 11 but where the coordinates for each observed point are not their values on variables 5 and 11 but

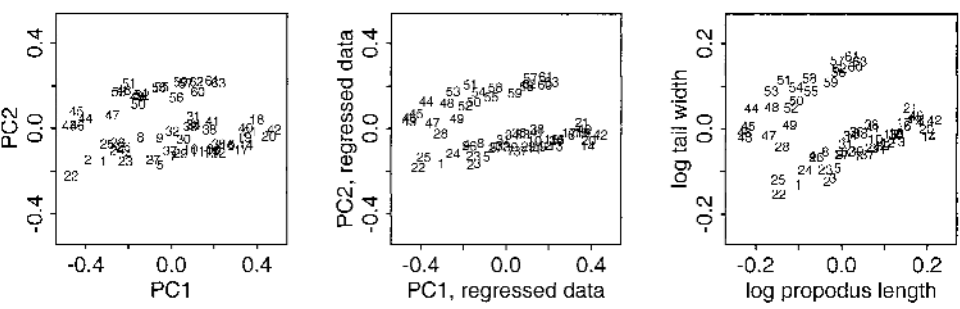


Figure 1. Scatterplot of 63 Crayfish on the First Principal Plane, i.e., the Best Two-Dimensional Approximation of the Full (Log-Transformed) Scatterplot (Left); Scatterplot of the Same (Log-Transformed) Crayfish Data on the Two Principal Components of the 13 Variables After Regression on Variables Tail Width and Propodus Length (Center); Scatterplot of the (Log-Transformed and Centered) Crayfish Data for the Variables Tail Width and Propodus Length (Right).

Table 2. Subsets of k Variables ($k = 2, 3, 4, 5$) of Somers' Crayfish Data and the Values of Their r_m and GCD Indicators. Superscripts are L for subsets chosen because their variables have the highest magnitude loading with one of the first k PCs; C for subsets chosen because they have the most correlated variable with one of the first k PCs; M for subsets chosen because they have the variables with the highest multiple correlation with the first k PCs; F for subsets chosen using forward selection (which, for the cases considered, are also the subsets chosen by stepwise selection). Subscripts of these superscripts are r as an indicator for r_m (2.8) and g as an indicator for GCD (2.5). Other subsets (without superscripts) are included because they are optimal for at least one of the indicators. The optimal value for each indicator for each cardinality appears in bold.

k	% Variance k PCs	Subset	r_m	% Variance (r_m^2)	GCD
2	79.92	$\{5, 13\}^L$	0.8352	69.75	0.8591
		$\{5, 11\}^{C, F_r, F_g}$	0.8697	75.63	0.9225
		$\{3, 11\}^M$	0.8633	74.53	0.8562
3	87.32	$\{5, 7, 13\}^L$	0.8703	75.75	0.7246
		$\{5, 7, 11\}^{C, F_r, F_g}$	0.9077	82.39	0.8233
		$\{3, 11, 12\}^M$	0.8974	80.52	0.7742
		$\{5, 7, 12\}$	0.8895	79.12	0.8761
		$\{5, 12, 13\}$	0.9111	83.00	0.8361
4	91.50	$\{5, 7, 12, 13\}^L$	0.9434	89.01	0.9594
		$\{5, 7, 11, 13\}^{C, F_g}$	0.9310	86.68	0.8908
		$\{7, 11, 12, 13\}^M$	0.8779	77.07	0.7520
		$\{5, 7, 11, 12\}^{F_r}$	0.9332	87.09	0.8801
5	94.12	$\{5, 6, 7, 12, 13\}^L$	0.9598	92.12	0.9520
		$\{5, 6, 7, 11, 13\}^{C, F_g}$	0.9478	89.84	0.8958
		$\{6, 7, 11, 12, 13\}^M$	0.9152	83.75	0.7929
		$\{5, 6, 7, 11, 12\}^{F_r}$	0.9499	90.24	0.8876

rather are their scores on the two relevant PCs for this two-dimensional subspace; hence, the difference from the third plot. The use of the criteria shows that a remarkable simplification is possible. The two-dimensional principal subspace can be closely reproduced by just two variables, thus greatly simplifying its original interpretation, in which the axes are defined by nontrivial loadings on many of the variables. Note that, if the choice of variables were to be based on variables with the highest magnitude loading on both PCs, as is often done, then a different subset of variables (variables 5 and 13) would be selected. This loadings-based subset is worse under both criteria (see Table 2 for details).

For three PCs, the percentage variance accounted for is 87.32%, and more than 95% of this optimum figure is the percentage variance accounted for by projecting the data onto the subspace spanned by only three (log-transformed) variables, i.e., tail width (variable 5), propodus width (variable 12), and dactyl length (variable 13), since these variables can, through regression, account for 83.00% of the data set's total variance. It should be pointed out that this three-variable optimum subset does not include variable 11 (propodus length), which is in the two-variable optimum subset. In addition, the three-variable subset

mentioned above does not optimize the GCD since it has a GCD of 0.8361, whereas the three variables 5, 7 (areola width), and 12 give rise to a GCD of 0.8761. This illustrates the fact that the two criteria do not necessarily produce the same optimum subset of the original variables, although experience with this and other examples together with the similarities between Equations (2.5) and (2.8) suggest that it is unlikely that subsets that are optimal for one criterion will perform poorly under the other.

We investigated whether stepwise search algorithms, as described by Neter et al. (1990, p. 453), would find, for this example, the best subsets according to our two criteria without looking at all subsets. For each criterion, the best single-variable subset was found. Then an iterative process was begun where, at each step and given a k -variable subset, an initial $(k + 1)$ -variable subset was determined by adding the variable (not in the k -variable subset) whose entry into the $(k + 1)$ -variable subset maximized the selected criterion. Before proceeding to $(k + 2)$ -variable subsets, a backward-type step was taken, where it was tested whether any of the k -variable subsets obtained by removing one of the variables already in the $(k + 1)$ -variable subset produced a higher value of the criterion than that that had been obtained with the k -variable subset. If so, that variable was removed, inducing a temporary return to k -variable subsets. A new forward-type step would then again incorporate a $(k + 1)$ th variable in the subset. If its value of the criterion exceeded that of the original $(k + 1)$ -variable subset, this new $(k + 1)$ -variable subset was chosen. Otherwise, the transition to $(k + 2)$ -variable subsets proceeded with the initially chosen $(k + 1)$ -variable subset.

This stepwise selection method (with a default forward direction) and a simple forward selection method were tested for both indicators. In Table 2, subsets chosen by several methods are compared for cardinalities $k = 2, 3, 4, 5$.

On the whole, this data set allows fairly good authentic dimensionality reduction, with k -variable subsets ($k = 1, \dots, 12$) behaving fairly similarly to the optimum k -PC subsets under both criteria. However, the optimum k -variable subsets are not necessarily contained in the optimum $(k + 1)$ -variable subsets. Algorithms that do not search all subsets do reasonably well for both criteria and, in any case, considerable effort at collecting and storing data can be spared with little loss of information as measured by either criterion.

4.2 THE PITPROP DATA

Jeffers' famous pitprop data set coincidentally also has 13 variables (originally used as regressor variables by Jeffers [1967]). Jeffers (1967) decided to retain the first six components, which account for some 87.0% of total variability, as can be seen in Table 3. He then attempted to interpret these six components, using all 13 variables in the interpretations (curiously, the use of variable 7 in the interpretation of the fourth PC seems to result from an error in transcribing the importance of the respective loading, i.e., for each PC, Jeffers retained variables whose loadings were at least 70% the magnitude of the largest loading and describes variable 7 as having 81% of the magnitude of variable 11's loading, whereas in reality its magnitude is only 8.1% of that of variable 11). However, the optimal six variables under the r_m criterion (2.8) (variables $\{x_2, x_4, x_5, x_7, x_{11}, x_{12}\}$) are sufficient to account

Table 3. Percentage Variance Accounted for by the First Six PCs in the Pitprop Data Set

<i>Principal component</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>
% Variance	32.5	18.3	14.4	8.5	7.0	6.3
Cumulative % variance	32.5	50.7	65.2	73.7	80.7	87.0

for 80.6% of the total variability and, with a seventh variable (x_8) added to this subset, we account for 86.6% of the total variability, i.e., 99.5% of the variability accounted for by the first six PCs.

The GCD indicator for the subspaces spanned by this seven-variable subset and by the first six PCs is 0.832 (the multiple correlations between each PC and all seven variables are greater than 0.96 for the first five PCs but are only 0.81 for the sixth PC). However, this indicator of subspace similarity can grow to 0.861 for a different subset of seven variables, i.e., variables 3, 5, 7, 8, 11, 12, and 13.

Jeffers (1967) uses the PC loadings of the variables to interpret the PCs, and such loadings are often taken, implicitly or explicitly, to suggest a subset of variables that can be used in a simplified interpretation. The seven-variable subset that is optimal for the r_m indicator (2.8) does not include three variables that would be chosen by most loadings-based methods of variable selection, i.e.,

- variable 3, which is the variable with the highest magnitude loading (0.541) for PC2,
- variable 13, which has (to three decimal places) the highest magnitude loading for PC6 ex aequo with variable 5 (a loading of 0.626),
- variable 1, which has the second highest magnitude loading for PC1 (0.404, compared with variable 2's loading of 0.406).

In addition, none of these three variables has the highest magnitude loading for any PC with rank greater than six, so that variable selection methods based on discarding variables with important loadings in the last few PCs would also fail to hit on this seven-variable subset.

At the same time, the optimal subset includes variables that would not be selected by loadings-based methods, i.e.,

- variable 8 has small magnitude loadings for all six main PCs (the loadings' magnitudes are, respectively, 0.294, 0.189, 0.243, 0.286, 0.185, 0.055; none of these loadings exceeds 75% of the largest loading for each PC and only one—for the first PC—clearly exceeds 50% of the largest loading); on the other hand, variable 8 has the second largest magnitude loading (0.642) for PC8;
- variable 4 does not have the greatest magnitude loading for any of the first six PCs but does so for PC12, with a loading of magnitude 0.585;
- variable 7 also does not have the greatest magnitude loading for any of the first six PCs but does so for PC11 (0.764).

Similar discrepancies exist between loadings-based subsets and those based on the GCD indicator (2.5). Fuller results are given in Table 4.

Table 4. Subsets of k Variables ($k = 2, 3, 4, 5, 6, 7$) of the Pitprop Data and the Values of Their r_m and GCD Indicators. Superscripts and subscripts have the same meaning as in Table 2, although superscript C is omitted since, for correlation matrix PCA, the largest magnitude loadings are equivalent to the largest magnitude correlations, and superscript S is added for subsets chosen using stepwise selection that differed from those chosen using forward selection.

k	% Variance k PCs	Subset	r_m	% Variance (r_m^2)	GCD
2	50.74	$\{2, 3\}^{L, F_g}$	0.6351	40.34	0.6954
		$\{1, 2\}^M$	0.5287	27.96	0.4163
		$\{2, 7\}^{F_r}$	0.6576	43.24	0.6241
		$\{3, 7\}$	0.6520	42.52	0.7598
		$\{1, 7\}$	0.6589	43.41	0.6515
3	65.19	$\{2, 3, 5\}^L$	0.7173	51.45	0.6994
		$\{1, 2, 7\}^M$	0.6725	45.23	0.5884
		$\{2, 3, 7\}^{F_g}$	0.7548	56.98	0.7943
		$\{2, 4, 7\}^{F_r}$	0.7605	57.84	0.8296
4	73.73	$\{2, 3, 5, 11\}^L$	0.7737	59.86	0.7245
		$\{1, 2, 4, 7\}^M$	0.7674	58.89	0.6326
		$\{2, 3, 7, 11\}^{F_g}$	0.8066	65.06	0.7917
		$\{2, 4, 7, 12\}^{F_r}$	0.8126	66.03	0.6919
		$\{2, 4, 7, 11\}$	0.8123	65.98	0.8195
5	80.73	$\{2, 3, 5, 11, 12\}^L$	0.8291	68.75	0.7962
		$\{2, 3, 4, 7, 11\}^M$	0.8367	70.01	0.7403
		$\{2, 3, 7, 11, 12\}^{F_g}$	0.8562	73.30	0.8584
		$\{2, 4, 7, 11, 12\}^{F_r}$	0.8613	74.18	0.8859
6	87.00	$\{2, 3, 5, 11, 12, 13\}^L$	0.8901	79.22	0.8927
		$\{1, 2, 3, 4, 7, 11\}^M$	0.8416	70.82	0.6865
		$\{2, 3, 5, 7, 11, 12\}^{F_g}$	0.8970	80.46	0.8615
		$\{2, 4, 5, 7, 11, 12\}^{F_r}$	0.8976	80.57	0.8598
7	91.43	$\{2, 3, 5, 9, 11, 12, 13\}^L$	0.9154	83.80	0.8573
		$\{2, 3, 4, 5, 7, 11, 12\}^M$	0.9018	81.32	0.8008
		$\{2, 4, 5, 7, 8, 11, 12\}^{F_r}$	0.9305	86.59	0.8778
		$\{2, 3, 5, 7, 11, 12, 13\}^{F_g}$	0.9266	85.87	0.8802
		$\{2, 3, 5, 6, 11, 12, 13\}^{S_g}$	0.9287	86.25	0.9046
		$\{3, 5, 7, 8, 11, 12, 13\}$	0.9197	84.58	0.9100

As a general comment, subset selections based on the traditional method of picking the (not previously selected) variable with the highest magnitude loading for each PC until some preassigned cardinality q (which in the tables of this paper is equal to the number of PCs retained, k) were found to be, at times, quite suboptimal (see Table 4, cardinalities 3, 4, and 5). The performance of the subsets of variables chosen because they were the q variables with the highest multiple correlations with the first k PCs seems even worse (again, for $q = k$, see all cardinalities shown in Table 4).

It should also be noted that the percentage of total variance accounted for by the first k

PCs can often be matched by a subset of just $k + 1$ variables. This is the significance of the square of the r_m criterion [Equation (2.8)], as discussed above. Values of r_m^2 can be seen in Tables 2 and 4.

4.3 PORTUGUESE FARM DATA

Data are available for Portuguese farms on 252 economic and agricultural variables within the framework of the European Union-wide Farm Accountancy Data Network. A subset of $n = 99$ farms and $p = 62$ variables was considered. The resulting (99×62) data set, along with the description of the 62 economic and agricultural variables that were retained, is available from the first author on request. Due to the diverse nature and measurement units of the $p = 62$ variables, a correlation matrix PCA was performed. As with many similar analyses, a fair number of PCs are necessary to account for even modest percentages of total variance. For example, 9 PCs are needed to account for 60% of the total variation and 16 PCs are required for 80%.

A referee has noted that separate analyses of economic and agricultural variables would be sensible. We agree, but the main reason for inclusion of this example is to demonstrate the practicality of using the criteria when there are large numbers of variables.

The (unit-norm) loadings for PCs generated by the correlation matrix for this data set are not given here for reasons of space. The large number of variables makes any attempt to interpret the first few PCs difficult unless a small number of variables that can approximate those PCs can be identified. It turns out that this can be done quite successfully.

For example, for three PCs, the percentage of total variance accounted for is 36.02. A loadings-based choice of three variables would choose variables 57 (gross production), 62 (return on capital), and 44 (forest surface area) to represent PCs 1, 2, and 3, respectively. This three-variable subset can account for 32.65% of total variance after regression (i.e., $r_m = 0.5714$). Indicator (2.5) is $GCD = 0.8095$ for this subset. Although this seems to be the optimal value for the GCD, a slightly better approximation can be achieved, in terms of indicator (2.8), with the three-variable subset $\{2, 57, 59\}$ (total surface area, gross production, and gross added value, respectively), for which $r_m = 0.5731$, i.e., for which 32.85% of total variance can be accounted.

As new PCs are added, variable subsets of the same cardinality can be found that provide good approximations to the information provided by those PCs. Examples of such subsets for $k = 10, 11, 12$ and the values of the indicators associated with them are given in Table 5. It can be seen that the suboptimality of the traditional loadings-based choice of subsets seems to get worse as the cardinality of the subsets grows. However, performing a complete search to identify the subset that optimizes any given criteria quickly becomes computationally prohibitive. For $k = 3$, there are 37,820 different three-variable subsets to test; for $k = 4$, this number rises to 557,845; for $k = 12$, we are already in the order of 2×10^{12} different subsets. The results of applying the forward and stepwise search methods to this data set are also included in Table 5. In most cases, the stepwise search produced the same result as the forward method. In those cases where there were differences, the

Table 5. Subsets of k Variables ($k = 10, 11, 12$) of the Portuguese Farm Data and the Values of Their r_m and GCD Indicators of the Quality of the Approximation, Which They Provide, Either to the Full Data Set or to Its First k PCs. Superscripts and subscripts are as in Tables 2 and 4 except for superscript M, which was not considered in this case.

% Variance			% Variance		
k	k PCs	Subset	r_m	(r_m^2)	GCD
10	65.39	$\{14, 23, 29, 30, 31, 44, 47, 49, 57, 62\}^L$	0.7472	55.83	0.7231
		$\{2, 12, 21, 30, 31, 39, 40, 46, 57, 61\}^{Fr}$	0.7707	59.39	0.7649
		$\{12, 15, 30, 31, 40, 44, 46, 49, 57, 62\}^{Fg}$	0.7647	58.48	0.7597
		$\{12, 30, 31, 38, 40, 44, 46, 49, 57, 62\}^{Sg}$	0.7651	58.53	0.7759
		$\{10, 11, 21, 30, 31, 39, 40, 44, 46, 58\}$	0.7716	59.53	0.7716
		$\{14, 25, 28, 30, 31, 39, 40, 44, 46, 50\}$	0.7646	58.46	0.7822
11	68.49	$\{14, 20, 23, 29, 30, 31, 44, 47, 49, 57, 62\}^L$	0.7638	58.34	0.7285
		$\{2, 12, 16, 21, 30, 31, 39, 40, 46, 57, 61\}^{Fr}$	0.7879	62.07	0.7553
		$\{12, 15, 20, 30, 31, 40, 44, 46, 49, 57, 62\}^{Fg}$	0.7808	60.96	0.7655
		$\{12, 20, 30, 31, 38, 40, 44, 46, 49, 57, 62\}^{Sg}$	0.7807	60.95	0.7749
		$\{14, 21, 29, 30, 31, 34, 39, 40, 44, 57, 59\}$	0.7903	62.46	0.7405
12	71.28	$\{14, 20, 23, 29, 30, 31, 34, 44, 47, 49, 57, 62\}^L$	0.7806	60.93	0.7222
		$\{2, 12, 16, 21, 30, 31, 34, 39, 40, 46, 57, 61\}^{Fr}$	0.8042	64.67	0.7697
		$\{9, 12, 15, 20, 30, 31, 40, 44, 46, 49, 57, 62\}^{Fg}$	0.7985	63.75	0.7663
		$\{9, 12, 20, 30, 31, 38, 40, 44, 46, 49, 57, 62\}^{Sg}$	0.7985	63.76	0.7747
		$\{9, 10, 12, 16, 25, 30, 39, 40, 44, 45, 46, 48\}$	0.8024	64.39	0.7937

resulting subsets are also included in Table 5. The subsets given in Table 5 that improve on the forward selection and stepwise searches were found by simulated annealing. This is an optimization technique that is less likely to get trapped in a local optimum than many other techniques (Aarts and Korst 1985). However, it is not guaranteed to find a global optimum and will not be readily available to many users of PCA.

5. CONCLUSIONS

The advantages of dimensionality reduction directly in terms of the original variables are clear—the data are more meaningful for the data analyst, data collection efforts may be spared in future studies, and underlying relations between the variables become more obvious. The disadvantages lie in the fact that it generates suboptimal lower dimensional representations (at least for the criteria that PCs optimize), the tidy break-up into uncorrelated components that PCA provides is not achieved, and the search for subsets that maximize any given criterion of good approximation is a computationally difficult problem.

The logical development of Section 2 leads to two indicators for proximity between a full data set or subsets of its PCs and subsets of the original variables. Both indicators are relatively simple and can be interpreted in terms of geometric concepts and in terms of standard statistical concepts. One of these indicators (the percentage of total variance

accounted for by the subset of the original variables) is maximized by one of McCabe's (1984, 1986) sets of principal variables. This relation with the matrix of partial covariances of the discarded variables, given the retained variables, used by McCabe (see also Appendix B) implies that this indicator is also related to the graphical modeling strategy and the RV-based strategy for variable subset selection, when standardized variables (correlation matrix PCA) are used, as is highlighted by the discussion in Falguerolles and Jmel (1993).

For the examples considered, suboptimality does not seem to be serious, in particular when the loadings-based selection of subsets is replaced with a selection method that explicitly seeks to maximize some criterion of good approximation. The stepwise selection algorithm, using either of these indicators as the criterion for inclusion/exclusion of variables in the subsets, seems to perform very reasonably, tending to produce near-optimal subsets without computationally prohibitive searching. A stepwise selection algorithm tends to perform better than the algorithm suggested by Gonzalez et al. (1990), which often resulted in the need for a full search of all possible subsets.

Similar behavior has been observed in a number of other data sets not presented here. These other examples also confirm that selection of variables based on loadings can be inadvisable, leading to quite different, and clearly suboptimal, choices of variables.

Finally, we note that we have considered PCA here in its usual exploratory role. We are interested in parsimoniously explaining the variability in the data set, with no thoughts of inference to a larger population. If such inference were relevant, then it might be desirable to incorporate some form of cross-validation when selecting an optimal subset.

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APPENDIX A

Let X be an $n \times p$, rank p , column-centered data matrix and $S = (1/n)X'X$ be its covariance matrix. Let the spectral decomposition of S be $S = \sum_{i=1}^p \lambda_i \mathbf{a}_i \mathbf{a}_i' = A \Lambda A'$ with Λ the diagonal $p \times p$ matrix of eigenvalues of S and A the orthogonal $p \times p$ matrix of eigenvectors of S . Given a set of q indices $\mathcal{G}$ (and its complementary set $\bar{\mathcal{G}}$), we have

$$S = \sum_{i \in \mathcal{G}} \lambda_i \mathbf{a}_i \mathbf{a}_i' + \sum_{i \notin \mathcal{G}} \lambda_i \mathbf{a}_i \mathbf{a}_i' = A_{\mathcal{G}} \Lambda_{\mathcal{G}} A_{\mathcal{G}}' + A_{\bar{\mathcal{G}}} \Lambda_{\bar{\mathcal{G}}} A_{\bar{\mathcal{G}}} = S_{\{\mathcal{G}\}} + S_{\{\bar{\mathcal{G}}\}},$$

where $A_{\mathcal{G}}$ and $\Lambda_{\mathcal{G}}$ are, respectively, the $p \times q$ and $q \times q$ matrices obtained by deleting from A all columns whose column number is not in set $\mathcal{G}$ and from Λ all rows/columns whose row/column numbers are not in $\mathcal{G}$. The matrices $A_{\bar{\mathcal{G}}}$ and $\Lambda_{\bar{\mathcal{G}}}$ are obtained likewise. Matrix $S_{\{\mathcal{G}\}} = A_{\mathcal{G}}\Lambda_{\mathcal{G}}A'_{\mathcal{G}}$ is a rank q $p \times p$ matrix. Its Moore–Penrose generalized inverse is given by $S_{\{\mathcal{G}\}}^{-} = A_{\mathcal{G}}\Lambda_{\mathcal{G}}^{-1}A'_{\mathcal{G}}$. We also have

$$S_{\{\mathcal{G}\}}^{-}S = S_{\{\mathcal{G}\}}^{-}(S_{\{\mathcal{G}\}} + S_{\{\bar{\mathcal{G}}\}}) = S_{\{\mathcal{G}\}}^{-}S_{\{\mathcal{G}\}} = A_{\mathcal{G}}\Lambda_{\mathcal{G}}^{-1}A'_{\mathcal{G}}A_{\mathcal{G}}\Lambda_{\mathcal{G}}A'_{\mathcal{G}} = A_{\mathcal{G}}A'_{\mathcal{G}} \quad (\text{A.1})$$

because $S_{\{\mathcal{G}\}}^{-}S_{\{\bar{\mathcal{G}}\}} = 0_{p \times p}$, since $A'_{\mathcal{G}}A_{\bar{\mathcal{G}}} = 0_{q \times (p-q)}$. In addition, and with similar reasoning, we have

$$SS_{\{\mathcal{G}\}}^{-}S = (S_{\{\mathcal{G}\}} + S_{\{\bar{\mathcal{G}}\}})S_{\{\mathcal{G}\}}^{-}S_{\{\mathcal{G}\}} = S_{\{\mathcal{G}\}}S_{\{\mathcal{G}\}}^{-}S_{\{\mathcal{G}\}} = S_{\{\mathcal{G}\}}.$$

Also, by direct substitution of (A.1), we have $XS_{\{\mathcal{G}\}}^{-}S = XA_{\mathcal{G}}A'_{\mathcal{G}}$.

APPENDIX B

McCabe's (1984) first three criteria all involve the matrix of partial covariances of the discarded variables (which we will call the subset $\bar{\mathcal{K}}$), given the retained variables (subset $\mathcal{K}$), i.e., the matrix $\Sigma_{\bar{\mathcal{K}}\bar{\mathcal{K}}|\mathcal{K}} = \Sigma_{\bar{\mathcal{K}}\bar{\mathcal{K}}} - \Sigma_{\bar{\mathcal{K}}\mathcal{K}}\Sigma_{\mathcal{K}\mathcal{K}}^{-1}\Sigma_{\mathcal{K}\bar{\mathcal{K}}} = (XI_{\bar{\mathcal{K}}})'(I - P_{\mathcal{K}})XI_{\bar{\mathcal{K}}}$, where $I_{\bar{\mathcal{K}}}$ is the submatrix of the $p \times p$ identity that results from deleting the k rows/columns associated with the variables in set $\mathcal{K}$. In the first criterion, McCabe minimizes the determinant of this matrix of partial covariances. The second criterion involves the minimization of the trace of $\Sigma_{\bar{\mathcal{K}}\bar{\mathcal{K}}|\mathcal{K}}$, and the third criterion involves the minimization of the trace of $\Sigma_{\bar{\mathcal{K}}\bar{\mathcal{K}}|\mathcal{K}}^2$. Now $\text{corr}(X, P_{\mathcal{K}}X)$ can also be written as

$$\begin{aligned} \text{corr}(X, P_{\mathcal{K}}X) &= \frac{\|P_{\mathcal{K}}X\|}{\|X\|} = \frac{\|X - (I - P_{\mathcal{K}})X\|}{\|X\|} = \sqrt{\frac{\text{tr}(X'P_{\mathcal{K}}X)}{\text{tr}(X'X)}} \\ &= \sqrt{\frac{\text{tr}[X'X - X'(I - P_{\mathcal{K}})X]}{\text{tr}(X'X)}} = \sqrt{\frac{\text{tr}(X'X) - \text{tr}(X'(I - P_{\mathcal{K}})X)}{\text{tr}(X'X)}} \\ &= \sqrt{1 - \frac{\text{tr}(X'(I - P_{\mathcal{K}})X)}{\text{tr}(X'X)}}. \end{aligned}$$

Maximizing this matrix correlation amounts to minimizing $\text{tr}(X'(I - P_{\mathcal{K}})X)$, but

$$\text{tr}(X'(I - P_{\mathcal{K}})X) = \sum_{i=1}^p x'_i(I - P_{\mathcal{K}})x_i = \sum_{i \notin \mathcal{K}} x'_i(I - P_{\mathcal{K}})x_i = \text{tr}((XI_{\bar{\mathcal{K}}})'(I - P_{\mathcal{K}})XI_{\bar{\mathcal{K}}}).$$

Hence, McCabe's second criterion is equivalent to maximizing Equation (2.8).